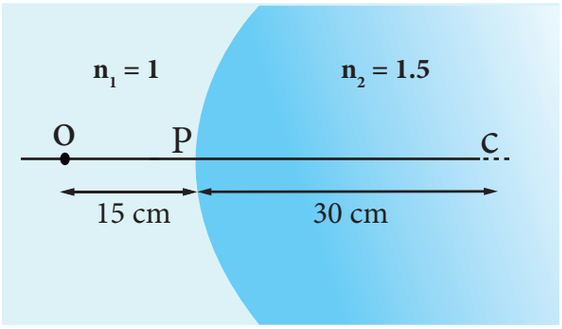
the centre of curvature of the separating surface.



Solution

Given, $u = -15$ cm, $R = 30$ cm, $n_1 = 1$ and $n_2 = 1.5$

Equation for single spherical surface is,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$$

Substituting the values,

$$\frac{1.5}{v} - \frac{1}{-15} = \frac{(1.5-1)}{30}; \qquad \frac{1.5}{v} + \frac{1}{15} = \frac{(0.5)}{30}$$

$$\frac{1.5}{v} + \frac{1}{15} = \frac{1}{60}; \qquad \frac{1.5}{v} = \frac{1}{60} - \frac{1}{15};$$

$$\frac{1.5}{v} = \frac{1-4}{60} = \frac{-3}{60}; = -\frac{1}{20}; \; v = -30 \text{ cm}$$

The image is a virtual image formed 30 cm to the left of the spherical surface.

6.6 THIN LENS

A lens is formed by a transparent material bounded between two spherical surfaces (or) one plane and another spherical surface. In a thin lens, the distance between the surfaces is very small. For two spherical surfaces, there will be two centres of curvature C_1 and C_2 and correspondingly two radii of curvature R_1 and R_2. For a plane surface, the centre of curvature C is at infinity and its radius of curvature R is infinity ($R = \infty$). The terminologies of spherical mirrors hold good very much for thin lenses except for the focal length.

6.6.1 Primary and secondary focus

As the thin lens is formed by two surfaces, the lens may separate two different media. i.e. the media to the left and right of the lens may be different. Hence, we have two focal lengths.

The *primary focus* F_1 is defined as a point where a point source kept produces a parallel emergent rays to the principal axis after passing through lens as shown in Figure 6.32. For a converging lens, such an object is a real object and for a diverging lens, it is a virtual object. The distance PF_1 is the *primary focal length* f_1.

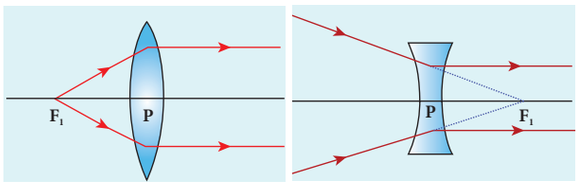


Figure 6.32 Primary focus

The *secondary focus* F_2 is defined as a point where all the parallel rays travelling close to the principal axis converge to form an image on the principal axis after passing through lens as shown in Figure 6.33. For a converging lens, such an image is a real image and for a diverging lens, it is a virtual image. The distance PF_2 is the secondary focal length f_2.

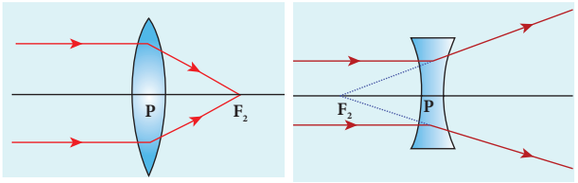


Figure 6.33 Secondary focus

UNIT-6(XII-Physics_Vol-2)_new.indd 29 21-07-2020 16:47:23